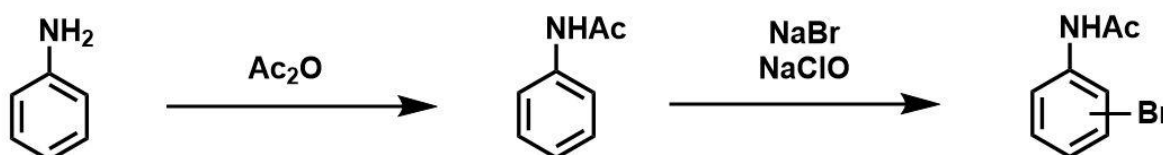


Activity 9: Electrophilic Aromatic Substitution Reactions: Bromination of Aniline and *N*-Phenylacetamide

The Experiment

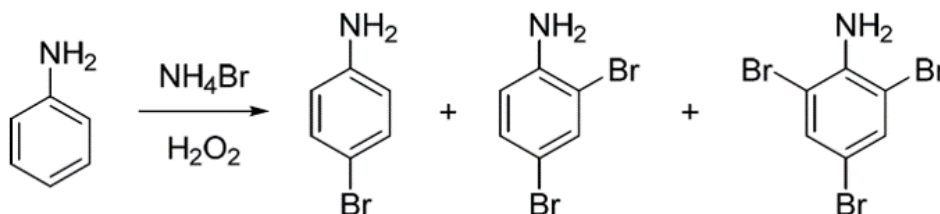
Electrophilic aromatic bromination of aniline with nitrogen protected as amide



Reference Experiment

- This reference experiment is not performed in lab but is used in the NMR analysis in the post-lab assessment.*

Electrophilic aromatic bromination of aniline (no nitrogen protecting group)



Lab Activity Goal

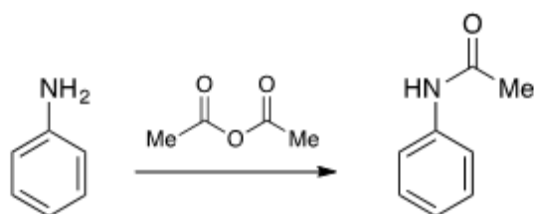
The electrophilic aromatic substitution of aromatic rings will be examined in the context of the electrophilic bromination of aniline and *N*-phenylacetamide. Specifically, the activating properties of amine substituents in electrophilic aromatic substitution reactions will be demonstrated. The effect of steric and electronic modifications to the aniline substrate will also be examined.

Pre-Laboratory Assignments

See the Activity 9 Pre-Lab Preparation and Outline document in Labflow.

Experimental Procedure

Part 1: Aniline to *N*-phenylacetamide



Note: Your instructor will dispense acetic anhydride reagent for you in your hood. While you are waiting for your instructor to “arrive”, you can prepare an ice bath and identify and find other equipment that will be needed.

1. Secure a 50 mL round bottom flask and stir bar to the monkey bars with a clamp. To this flask your instructor will add 2 mL of acetic anhydride. Stopper the flask.
2. Lower the stoppered flask into an ice water bath (water is necessary for good thermal transfer from reaction flask to ice mixture) that is on top of a stir plate (plugged into an outlet, not the Variac).

- **Note:** Obtain two 1 mL syringes (green color), place them in a 25 mL Erlenmeyer flask and nest this flask in a 400 mL beaker. Take this assembly to the aniline dispensing hood.



Important: Safety Pause

You will need to wear LATEX gloves to handle aniline. If you have a latex allergy, then someone else will need to do this step.

Safety Pause

Activity 9: Procedure

3. Bring your syringe assembly to the aniline dispensing hood. From a stock beaker of aniline (prepared by your instructor), draw up 0.5 mL of aniline into each syringe.
 - **Note:** Do not draw 1 mL up into a single syringe. The plunger can easily come loose from the syringe barrel at this volume.
4. Carefully take your aniline-loaded syringe assembly back to your hood. Record the exact volume in your syringes so that you can calculate theoretical yield.
5. Add the aniline dropwise to the acetic anhydride over 5 minutes with stirring.
 - **Note 1:** The reaction mixture will solidify as the reaction proceeds. If necessary, break up the solid with a spatula so that you can continue to stir the reaction.
 - **Note 2:** Once you have added all your aniline, clean the syringes as demonstrated by your instructor and explained below.

Cleaning aniline syringes: In the aniline dispensing hood while wearing latex gloves, draw clean acetone into each syringe from the labeled beaker or Erlenmeyer flask. Empty the aniline/acetone wash into the specially labeled aniline waste beaker or flask. The rinsed syringes should then be placed into the red solid waste. Remove your latex gloves and throw them into the red solid waste as well. For the rest of the lab, nitrile gloves are appropriate.

6. Add 10 mL of 0.5 M NaOH solution to the reaction flask. Heat this solution over a steam bath for 1 minute to dissolve the solid enough so that solid is no longer sticking to the sides of the flask. Pour the reaction mixture into a 125 mL Erlenmeyer flask. Rinse any solids remaining in your round bottom reaction flask into the Erlenmeyer using 10 mL of 0.5 M NaOH. Repeat this rinsing with a second portion of 10 mL of 0.5 M NaOH.

Activity 9: Procedure

- Let the mixture cool to ambient temperature then place the flask in an ice bath (~10 minutes). Collect the resulting colorless precipitate by vacuum filtration. Dispose of the filtrate in the designated aqueous waste.
- Reserve a **small** (IR size) sample of the solid product (*N*-phenylacetamide) and obtain an IR spectrum of your Part 1 product. *Use the remaining product in Part 2.*

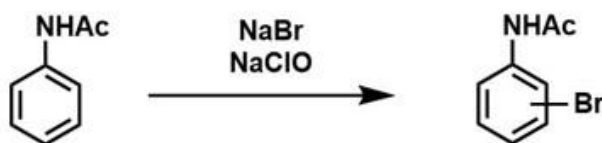


Important: Safety Pause

CAUTION: Be sure that all aniline has been removed from all laboratory hoods and benches **before proceeding** to Part 2. It is very important that the bleach solution used in Part 2 **never** comes into contact with aniline. Your instructor **must confirm** that this has been done. Sodium hypochlorite (bleach) is a strong oxidant and can release noxious gas.

Safety Pause

Part 2: Bromination of *N*-Phenylacetamide



- Transfer the solid product obtained in Part 1 to a 125 mL Erlenmeyer flask equipped with a stir bar.
- Obtain 10 mL 95% ethanol and 10 mL of glacial acetic acid from the dispensing station.
 - For **each** reagent, bring an appropriate graduated cylinder nested inside a large beaker capped with an inverted smaller beaker to the dispensing station. This method should be your standard method for obtaining liquids from the TA chemical hood.
 - Transfer the appropriate amount of each reagent (from the labelled stock beakers) to the graduated cylinder using a plastic pipette. Be sure the pipette is always vertical (pointed straight down). This

Activity 9: Procedure

prevents increasing vapor pressure in the bulb from pushing a reagent stream out of the pipette in undesirable directions (onto you or others in the lab).



Important: Safety Pause

CAUTION: Acetic acid is corrosive. Contact with eyes, skin or clothing should be rigorously avoided. Breathing the vapors should also be avoided. Don't pour directly from the reagent bottle into a small graduated cylinder.

Safety Pause

3. Add the ethanol and acetic acid to your Erlenmeyer flask that contains the acetamide intermediate. Secure the flask to the monkey bars and lower into an ice bath on the stir plate (plugged into the outlet not the variac).
4. Obtain ~2.8 g of sodium bromide. Add NaBr to the acetamide solution. Stir the reaction for 5 min at 0 °C. Clean up any mess you make at the balance area.



Important: Safety Pause

CAUTION: Be sure the bleach solution that you are about to begin working with is **not mixed with acetone**. If you have just cleaned your graduated cylinder for step 5, it must be rinsed with water before using it to measure bleach.

Safety Pause

5. Measure 18 mL of bleach into an appropriate graduated cylinder. You can bring the bottle of household aqueous bleach to your hood for the measurement. Return the bleach to the TA chemical hood when you are finished.
6. Add the aqueous bleach solution to the reaction solution. Stir the reaction for 60 seconds then remove the ice bath and keep stirring the reaction at room temperature for another 15 minutes.

Activity 9: Procedure

7. After stirring the reaction for 15 minutes, cool the reaction mixture in an ice bath. With continued stirring with the reaction mixture submerged in the ice bath, add 2 g sodium thiosulfate ($\text{Na}_2\text{S}_2\text{O}_3$; to quench excess bromine) to the reaction flask. Grind 2 g NaOH using a mortar and pestle. Add the ground NaOH to the reaction flask (to fully precipitate the reaction product). Remember to clean up any mess you make at the balance area.
8. Continue stirring the reaction for an additional 5 minutes to maximize dissolution of the $\text{Na}_2\text{S}_2\text{O}_3$ and NaOH. Note: all the solid may or may not dissolve.
 - **Note:** The reaction product will not be soluble, so the reaction mixture will **not** become homogeneous.
9. Vacuum filter the precipitated product. Rinse the product with cold 50% aqueous ethanol (2 x 5 mL) then allow the product to air dry while continuing to pull vacuum for 2-3 minutes. Collect the rinsed product and dispose of the filtrate in the designated aqueous waste.

Part 3. Purification

1. Add a couple of boiling stones to ~50 mL of 50% aqueous ethanol. Heat on a steam bath.
2. Transfer the solid from Step 9 into another 125 mL Erlenmeyer flask and purify the crude product by recrystallization using 50% aqueous ethanol as the solvent.
 - **Note:** Be careful not to add too much solvent.
3. Collect the crystallized product by vacuum filtration and allow the product to air dry.

Part 4. Spectroscopic Analysis

Activity 9: Procedure

Once your sample is dried, weigh it and then obtain an IR spectrum of your Part 2 product. Look for absorptions consistent with the acetyl group, the aromatic group, and aromatic substitution.

Before you leave lab, upload your in-lab notebook pages to Labflow. Your in-lab notebook pages must include the following in a **single pdf file**:

- Any experimental observations (including but not limited to: color changes, observational notes, measurements, sketches/images of your TLC plates, etc.)
- Your handwritten answers to the questions posted in the In-Lab Notebook Pages tile on Labflow
- A picture of your Organic Lab Cleaning Checklist

Clean-Up and Waste Disposal

Activity 9 Specific Clean-Up

- **Organic Liquid Waste:**
 - Acetone rinse from glassware washing, ethanol/water mixtures (but NOT aniline)
 - Excess ethanol (if any)
- **Aqueous Waste:**
 - **NO ACETONE!**
 - Any **aqueous** glassware rinses (rinsed with water, not acetone)
 - The filtrate from part 1, step 7
 - Aqueous NaOH, bleach, acetic acid, filtrate from part 2, step 9.
 - Excess acetic anhydride (if any)
- **Aniline Special Waste:**
 - Acetone rinses from aniline syringes
- **Solid Waste:**
 - Solid reaction products
 - excess NaBr, sodium thiosulfate, and solid NaOH

Activity 9: Procedure

- Rinsed syringe barrels (note that if our syringes had needles, the needles would need to go in a "sharps" waste)
- Used filter paper
- Used gloves (latex and nitrile)

Routine Clean-Up

- Clean up your hood space of all trash, spilled chemicals, etc. Don't leave a mess for the people who share this hood at other time slots. Up to 24 other people are sharing this space at various time slots throughout the week. Clean up your mess!
- No dirty or clean glassware should be left in the sink. No used filter paper or wadded up dirty paper towels should be left in the sink.
- **Dirty paper towels go into the REGULAR TRASH** not the red solid waste trash. This saves the university a lot of waste disposal money!
- Dispose of dirty filter paper in the red solid waste container.
 - Do **NOT** deposit filter paper in the liquid waste eco-funnel as this will cause a clog which will lead to a messy overflow of waste .
 - Do **NOT** leave filter paper in the sink as this is poor lab etiquette and rude.
- Used gloves should be disposed of into the red solid waste.
- **Ask your instructor if you are in charge of cleaning up the balance area** for this week. If you are in charge do the following:
 - Be sure all reagent bottles are properly capped.
 - Use the small brush to clean off balances.
 - Use small broom and dustpan (located on the wall surrounding the balance area) to remove excess solid from counter. Dispose of excess solids in red solid waste container. Return broom and dustpan to proper location.
 - Wipe down the counter with a moist paper towel.
- **Ask your instructor if you are in charge of making sure the sink area is presentable** for the next lab group.
 - Wash any dirty glassware rudely left in the sink. Let clean glassware air dry on rack or window sill area.

Activity 9: Procedure

- Dispose of any paper towels, filter paper, or cotton rudely left in the sink.
- **Do not remove broken glass from the drain or the sink.** Rather, tell your instructor and then fill out a "something wrong form" located in the cabinet above the solid waste containers. Be sure to record room number and the Left/Right location of the sink.

Post-Lab Assignment

The Activity 9 Post-Lab Assessment is posted on Labflow and is due according to the deadline posted on Labflow.

References

1. Cardinal, P.; Greer, B.; Luong, H.; Tyagunova, Y. *J. Chem. Educ.* **2012**, 89, 1061–1063.